FairLight: City-Wide Emergency Vehicle Route Prioritization via Envy-Free Hierarchical Multi-Agent Actor–Critic

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Abstract

Urban traffic signal control is a complex sequential decision-making problem in which decisions made at one intersection can influence traffic conditions at neighboring and downstream intersections. The problem becomes increasingly challenging as the size of the road network grows because a controller must simultaneously achieve network-level coordination, local responsiveness, and computational scalability. Reinforcement learning has emerged as a promising approach for adaptive traffic signal control, while multi-agent reinforcement learning (MARL) allows control to be distributed among individual intersections. However, fully decentralized approaches may lack sufficient global context, whereas centralized approaches become difficult to scale.

This paper proposes **FairLight**, a **Hierarchical Region-Aware Multi-Agent Reinforcement Learning framework for Fair and Adaptive Traffic Signal Control**. The framework introduces a regional abstraction between network-level coordination and intersection-level decision-making. The traffic network is partitioned using graph community detection. Regional traffic states are subsequently processed by a high-level Meta-Policy using spatial and temporal representations, while low-level intersection agents use local observations, neighboring information, and regional guidance to select signal actions.

A central feature of the proposed framework is the explicit treatment of **fairness as a traffic-control objective**. Conventional evaluation based only on average delay, average travel time, or throughput can hide excessive delays experienced by a smaller group of vehicles. The proposed framework therefore incorporates vehicle-level waiting-time distribution, maximum waiting time, tail waiting time, and Jain's fairness index into optimization and evaluation. A multi-objective reward function jointly considers traffic efficiency, fairness, and emergency responsiveness.

The framework also treats **region construction as part of the research problem rather than merely a preprocessing step**. DBSCAN, Louvain, Leiden, and hybrid region construction methods are investigated to determine how the choice of regional abstraction affects downstream traffic-control performance.

Keywords: Traffic Signal Control, Multi-Agent Reinforcement Learning, Emergency Vehicle Routing, Community Detection, Louvain Algorithm, Fairness-Aware Optimization, Hierarchical Reinforcement Learning

1 Introduction

Urban traffic signal control is a complex sequential decision-making problem where decisions at one intersection propagate to neighboring and downstream intersections. Traffic demand varies dynamically with time, incidents, weather, and travel patterns, making static or rule-based controllers insufficient for real-world conditions. Adaptive systems such as SCOOT introduced traffic-responsive coordination [2], while reinforcement learning approaches have since enabled data-driven signal control [11]. Multi-agent reinforcement learning (MARL) further distributes control across intersections, improving scalability, though fully decentralized approaches risk optimizing locally at the expense of network-wide coordination [20, 19].

Beyond aggregate congestion, traffic signal decisions simultaneously affect travel time, delay, queue formation, throughput, fuel consumption, and emissions. Critically, evaluating controllers solely through aggregate metrics such as average waiting time conceals how service is distributed among individual vehicles. Consider two controllers with vehicle waiting times $W_A = [10, 10, 10, 10, 10, 100]$ and $W_B = [25, 25, 25, 25, 25, 25]$: Controller A achieves a lower mean delay yet inflicts severe starvation on one vehicle. This motivates fairness as a first-class control objective rather than a post-hoc evaluation criterion.

A further dimension of real-world uncertainty that existing MARL frameworks largely overlook is the presence of emergency vehicles (EMVs). Most traffic signal control approaches, including hierarchical methods such as HiLight [24] and cooperative approaches such as CoLight [17], assume homogeneous vehicle populations and optimize purely for aggregate efficiency. In practice, ambulances, fire engines, and police vehicles impose stochastic, high-priority demands on the network that cannot be resolved by ordinary signal timing. Delayed EMV passage has measurable consequences: studies report approximately 700 annual fatalities attributable to late ambulance responses, and an estimated annual mean of 4,500 ambulance-involved crashes in the United States [35]. EMVLight demonstrated that jointly optimizing EMV routing and adaptive signal control within a multi-agent actor-critic framework substantially reduces emergency response times [31], but does not address hierarchical city-wide coordination or fairness. These considerations establish EMV responsiveness as a necessary component of any realistic traffic management framework, not an optional extension. At the same time, providing unrestricted priority to an EMV may excessively delay ordinary traffic, potentially transferring congestion to other parts of the network rather than resolving it. EMV prioritization should therefore not be treated as an isolated intersection-level problem: it requires coordination across the surrounding traffic network while maintaining acceptable service for non-emergency vehicles.

Hierarchical reinforcement learning addresses the scalability–coordination tradeoff by separating long-horizon strategic decisions from short-horizon local control [22, 23]. HiLight is particularly relevant here, employing a Meta-Policy with Transformer-based spatial and LSTM-based temporal modelling to coordinate decentralized Sub-Policy agents across large networks [24]. However, its region partitioning is treated as a preprocessing artifact rather than a control-relevant variable, and it neither incorporates EMV priority nor vehicle-level fairness.

This paper proposes **FairLight: City-Wide Emergency Vehicle Route Prioritization**, an **Envy-Free Hierarchical Multi-Agent Actor-Critic** framework that jointly addresses these limitations. The framework partitions the traffic network using graph community detection methods (DBSCAN, Louvain, Leiden, and hybrid variants), treating region construction as a research variable whose effect on downstream control performance is explicitly investigated. A hierarchical coordination layer issues regional guidance to local intersection agents, while a multi-objective reward function simultaneously optimizes traffic efficiency, vehicle-level fairness, and EMV responsiveness. To the best of our knowledge, no existing framework jointly integrates hierarchical MARL, topology-aware regional partitioning, envy-free fairness optimization, and dynamic multi-EMV routing within a unified city-wide architecture.

2 Related Work

Conventional traffic signal control has primarily relied on analytical optimization and rule-based methods for determining optimal signal timings. Webster's signal timing formulation introduced one of the earliest mathematical models for fixed-cycle traffic signal optimization, while later adaptive traffic control systems such as SCOOT and SCATS dynamically adjusted signal timings using detector-based traffic measurements and predefined control strategies [1, 2, 3, 4]. Max-pressure control further improved network throughput by selecting signal phases according to queue pressure differences between incoming and outgoing links [5]. Although these approaches are computationally efficient and have been successfully deployed in practice, they rely heavily on predefined traffic models and handcrafted optimization rules. Consequently, they struggle to adapt to highly dynamic urban environments involving unpredictable congestion, heterogeneous intersections, accidents, and emergency vehicle requests.

To improve adaptability, several optimization and machine learning techniques have been introduced for traffic signal control. Evolutionary optimization methods such as Genetic Algorithms optimize signal timing plans through iterative search, while Fuzzy Logic Controllers employ expert-defined rules to respond to uncertain traffic conditions [6, 7]. Later studies adopted machine learning techniques including Support Vector Machines, Artificial Neural Networks, and Long Short-Term Memory (LSTM) networks to predict traffic flow and congestion using historical observations [8, 9, 10]. Although these methods improve prediction accuracy and optimization quality, they primarily estimate future traffic states rather than learning sequential traffic control policies. Furthermore, they require extensive historical training data and cannot continuously adapt to rapidly changing traffic conditions.

Recent advances in deep reinforcement learning have significantly improved adaptive traffic signal control

by enabling traffic controllers to learn directly through interaction with the traffic environment. IntelliLight demonstrated the effectiveness of Deep Q-Networks for adaptive traffic signal control at isolated intersections [11]. PressLight integrated max-pressure traffic theory into deep reinforcement learning to maximize traffic throughput while reducing handcrafted reward engineering [12]. FRAP modeled traffic control as a phase competition problem and exploited traffic phase symmetry to improve learning efficiency [13]. AttendLight later introduced an attention-based architecture capable of generalizing across heterogeneous intersections with varying lane configurations [14]. MetaLight focused on transferring learned policies across unseen intersections [15], while Advanced-XLight demonstrated that effective traffic state representation can significantly influence performance in reinforcement-learning-based traffic signal control [16]. Despite these advances, existing deep reinforcement learning methods primarily optimize local traffic efficiency and do not jointly consider emergency vehicle prioritization, fairness, or large-scale hierarchical coordination.

To improve cooperation among multiple intersections, graph neural networks and multi-agent reinforcement learning have received increasing attention. CoLight introduced graph attention networks to enable communication among neighboring intersections for cooperative traffic signal control [17]. MPLight further improved scalability by introducing parameter sharing across reinforcement learning agents, allowing efficient deployment over city-scale transportation networks [18]. HiLight extended this concept by proposing a hierarchical reinforcement learning framework consisting of Meta-Policy and Sub-Policy agents for coordinating thousands of intersections efficiently [24]. CityFlow provides a large-scale multi-agent reinforcement learning environment designed specifically for city traffic scenarios and supports evaluation of scalable traffic-control algorithms [21]. Although these methods substantially improve scalability and inter-agent cooperation, their primary objective remains maximizing traffic efficiency. They neither address emergency vehicle routing nor incorporate fairness-aware resource allocation across different regions of a city.

HiLight's hierarchical Meta-Policy/Sub-Policy formulation [24], like most of the methods discussed above, operates under a closed-world assumption: it is trained and evaluated on recurrent congestion patterns and does not model non-recurrent disruptions such as emergency vehicle preemption or traffic incidents. This omission is not merely conceptual. Poorly coordinated EMV preemption has been shown to increase average delay for cross-traffic by 18–32 seconds per vehicle per preemption event, with these localized delays cascading into network-wide congestion [32]. Even well-established preemption strategies carry a measurable efficiency cost: EMVLight's own evaluation shows that traditional preemption techniques increase average non-EMV travel time by approximately 10% relative to unpreempted baselines [31]. These results indicate that emergency vehicle disruption is a first-order cost of operating a real traffic network, not a negligible edge case that a closed-world controller can safely ignore. Motivated by this, our framework treats EMVs and accident events as first-class constraints on the control problem rather than external events layered on top of it, at the cost of an increase in average travel time and delay relative to a disruption-free benchmark—a cost we treat as the price of a tighter approximation to real-world operating conditions.

Emergency vehicle routing and traffic signal pre-emption have also received considerable research attention. Conventional emergency routing methods generally employ shortest-path algorithms such as Dijkstra's algorithm, A*, and various time-dependent shortest path formulations to determine optimal routes under static or partially dynamic traffic conditions [25, 26, 27, 28, 29]. More recent studies have introduced reinforcement learning-based routing strategies capable of adapting to changing traffic environments [30]. EMVLight jointly optimizes emergency vehicle routing and adaptive traffic signal control using multi-agent actor–critic reinforcement learning, significantly reducing emergency response times while maintaining acceptable traffic efficiency for ordinary vehicles [31]. Subsequently, hierarchical multi-agent actor–critic methods have been proposed to further improve emergency routing through coordinated traffic signal control [33]. However, existing emergency traffic management approaches generally assume decentralized coordination and do not explicitly consider fairness, topology-aware regional decomposition, or simultaneous optimization for multiple emergency vehicles operating within large urban transportation networks. Visual sensing-based approaches have further demonstrated that incorporating real-time distance measurement and vehicle counting into traffic management can substantially improve EMV scheduling and intersection clearance [35].

Fairness has recently emerged as an important objective in intelligent transportation systems. FairMARL introduced a fairness-aware multi-agent reinforcement learning framework based on generalized social welfare functions to reduce disparities among intersections while maintaining competitive traffic efficiency [34]. Although fairness-aware optimization improves equitable resource allocation, existing approaches primarily

focus on balancing intersection delays and do not integrate emergency vehicle routing, hierarchical coordination, or topology-aware traffic management into a unified framework.

Overall, significant progress has been achieved in analytical optimization, deep reinforcement learning, graph-based coordination, emergency vehicle routing, and fairness-aware traffic signal control. However, these objectives have largely been investigated independently. Classical optimization methods lack adaptability, deep reinforcement learning approaches prioritize traffic efficiency over fairness, graph-based coordination methods do not explicitly support emergency response, emergency vehicle routing methods overlook hierarchical city-wide coordination, and fairness-aware methods ignore emergency traffic management.

To the best of our knowledge, no existing framework jointly integrates **hierarchical multi-agent reinforcement learning, topology-aware regional partitioning, envy-free fairness optimization, and dynamic multi-emergency vehicle routing** within a unified city-wide traffic management architecture. The proposed framework, **FairLight**, addresses these limitations by introducing an **Envy-Free Hierarchical Multi-Agent Actor–Critic** architecture that simultaneously optimizes traffic efficiency, fairness, emergency response, and scalable coordination across heterogeneous urban transportation networks.

3 Proposed Framework

The proposed framework, **FairLight**, is organized around three complementary methods that together address scalability, emergency responsiveness, and equity in city-wide traffic signal control. First, the road network is automatically partitioned into coherent subregions using graph-based community detection (the Louvain algorithm), removing the need for manual, expert-driven region design and allowing the hierarchical Meta-Policy/Sub-Policy architecture to scale to arbitrarily large networks. Second, the framework extends single-vehicle emergency routing to a **multi-EMV** setting, in which several emergency vehicles may simultaneously request priority across overlapping parts of the network, requiring explicit conflict resolution and coordinated corridor clearing. Third, the framework treats **fairness** and **accident-induced congestion** as first-class control objectives rather than side effects, so that the learned policy neither systematically starves individual approaches nor collapses when unexpected incidents disrupt normal traffic flow. The remainder of this section describes each of these three methods in turn.

3.1 Clustering Based on Louvain

In the proposed framework, manually dividing the road network into precise subsections is impractical for real-world scenarios due to the large number of roads and intersections involved. Therefore, the road network is modeled as a weighted graph $G = (V, E)$, where intersections are represented as nodes and road segments as edges.

To automatically identify suitable subsections, the **Louvain community detection algorithm** is applied to the graph. The algorithm partitions the network into communities by maximizing **modularity**, which measures the difference between the observed connectivity within communities and the connectivity expected in a corresponding random network:

$$Q = \frac{1}{2m} \sum_{i,j} \left(A_{ij} - \frac{k_i k_j}{2m} \right) \delta(c_i, c_j), \tag{1}$$

where A_{ij} is the edge weight between nodes i and j, k_i and k_j are their weighted degrees, m is the total edge weight, and $\delta(c_i, c_j)$ indicates whether the nodes belong to the same community.

A higher modularity indicates stronger community structure, allowing the identified communities to serve as automatically generated road-network subsections. Each subsection is assigned a **sub-policy**, while a higher-level **meta-policy** coordinates the behavior of these sub-policies across the complete network. This provides a scalable alternative to manual decomposition and enables hierarchical policy optimization over complex road networks.

3.2 Multiple EMV Vehicle Routing Across the City

Prior emergency-vehicle (EMV) prioritization approaches in the traffic-signal-control literature, including the single-agent preemption strategies discussed in Section 2, are almost exclusively formulated around a

single emergency vehicle traversing the network at any given time. This assumption breaks down in dense urban environments, where it is common for multiple ambulances, fire trucks, and police vehicles to be simultaneously active, often converging on overlapping segments of the road network from different origins and with different destinations. Extending EMV prioritization to a genuinely multi-vehicle setting therefore requires the framework to reason jointly about several time-critical routes rather than optimizing each one in isolation.

We propose to model the multi-EMV routing problem as a joint optimization over a set of active emergency vehicles $\mathcal{E} = \{e_1, e_2, \ldots, e_K\}$, each associated with a current position, a destination, and a priority class (for example, life-threatening medical emergency versus structural fire response). At each decision step, the framework must determine (i) a preferred route for every active emergency vehicle through the signalized network, and (ii) how to allocate green-phase priority at any intersection that lies on more than one emergency vehicle's preferred route. This second requirement is the central new difficulty relative to single-EMV routing: when two emergency vehicles approach the same intersection from conflicting directions within a short time window, granting unconditional priority to both is physically impossible, and the controller must instead arbitrate based on factors such as estimated time-to-intersection, remaining distance to destination, and priority class.

We envision extending the existing Meta-Policy/Sub-Policy hierarchy so that each active emergency vehicle contributes an additional, time-decaying bias term to the regional and intersection-level state representations along its anticipated route, rather than introducing a wholly separate routing module. Conceptually, this bias term would increase in magnitude as an emergency vehicle approaches a given intersection and decay once it has passed, so that the learned Sub-Policy naturally favors clearing a path ahead of an approaching emergency vehicle while reverting to its ordinary efficiency/fairness trade-off once the vehicle has moved on. Coordination across subregions, needed when an emergency vehicle's route crosses several Louvain-derived subregions in sequence, would be handled by the Meta-Policy, which is already responsible for issuing cross-regional guidance.

Resolving conflicts between multiple simultaneously active emergency vehicles is expected to require an explicit arbitration rule layered on top of this bias mechanism — for instance, prioritizing the vehicle with the least estimated time-to-intersection, breaking ties by priority class, and bounding the maximum delay that can be imposed on a lower-priority emergency vehicle so that no single vehicle is perpetually deprioritized in favor of others. The corresponding evaluation methodology would extend the single-EMV response-time metrics used elsewhere in this work (e.g., average and 95th-percentile emergency waiting time) to the multi-vehicle case by additionally reporting the spread of response-time improvements across all active emergency vehicles, so that the framework can be shown to accelerate every emergency vehicle's passage and not merely the one considered by the controller to be of highest priority.

Finally, we note that a multi-EMV setting also changes how the framework's fairness objective (Section 3.3) should be interpreted: granting priority to emergency vehicles necessarily imposes additional waiting time on ordinary traffic, and this additional cost must be distributed across the network rather than concentrated on whichever ordinary vehicles happen to be sharing a road segment with an active emergency-vehicle corridor at a given moment. This interaction between multi-EMV routing and the fairness reward is left as an open design question to be resolved in the full version of this section.

3.3 Fairness and Accidental Congestion Control

Traffic-signal controllers optimized purely for network-wide efficiency metrics, such as average travel time or average delay, can produce policies that are efficient in aggregate yet highly inequitable at the level of individual approaches or vehicles. A controller may, for instance, learn to sacrifice one chronically low-priority approach in order to keep throughput high on the network's dominant corridors, resulting in a small subset of vehicles experiencing persistent, disproportionate waiting times even while the network-wide average delay looks acceptable. The fairness component of the proposed framework is intended to make this trade-off explicit and controllable rather than an incidental by-product of reward design.

We plan to quantify fairness along several complementary axes: the variance of waiting times across approaches at a given intersection, the maximum (tail) waiting time experienced by any single approach, and Jain's Fairness Index computed over per-approach or per-vehicle service levels. Unlike a single scalar efficiency term, these measures are each sensitive to a different failure mode — variance captures persistent

imbalance between approaches, the tail-waiting-time penalty specifically targets starvation of the worst-served approach, and Jain's index provides a normalized, scale-independent summary of overall equity that can be tracked and compared across scenarios and partitioning methods. An "envy" term, comparing each intersection's service level against that of its neighboring intersections, is further intended to discourage locally greedy policies that improve one intersection's performance strictly at a neighbor's expense — directly relevant to the Louvain-based regional decomposition of Section 3.1, since poorly balanced subregions could otherwise induce exactly this kind of neighbor-level inequity.

A second, largely orthogonal concern addressed by this subsection is robustness to **accidental congestion**: collisions, breakdowns, and other unplanned incidents that suddenly and sharply reduce the effective capacity of one or more road segments. Conventional traffic signal controllers are typically trained and evaluated exclusively under nominal demand conditions, and so may respond poorly when a lane or approach is unexpectedly blocked, continuing to allocate green time as though the obstructed segment retained its usual capacity. We intend to address this by exposing the controller to a representative distribution of injected incidents during training — varying in location, duration, and severity — so that the learned Sub-Policy and Meta-Policy learn to detect the resulting shift in local traffic statistics (e.g., a sudden rise in queue length or a collapse in mean speed on an otherwise normal approach) and reallocate green time accordingly, rather than requiring a separate, hand-engineered incident-detection and override layer.

We additionally plan to evaluate whether accident-aware training degrades the framework's fairness guarantees: an incident response that simply redirects all available green time toward the obstructed approach could, in principle, starve unrelated approaches elsewhere in the network. The evaluation methodology for this subsection will therefore report fairness metrics (variance of waiting time, tail waiting time, Jain's index) separately for incident and incident-free simulation intervals, to confirm that the controller's response to accidental congestion does not come at the cost of the equity objectives established earlier in this section. Together with the multi-EMV routing extension of Section 3.2, this completes the three methods — scalable graph-based regional decomposition, coordinated multi-emergency-vehicle routing, and fairness-aware, incident-robust congestion control — that constitute the proposed FairLight framework.

3.4 Hierarchical Architecture Overview

The proposed FairLight framework addresses the scalability and coordination challenges of city-scale adaptive traffic signal control through a hierarchical multi-agent reinforcement learning architecture. A fully centralized controller becomes computationally expensive as the number of intersections increases, whereas independent intersection-level agents lack sufficient network-wide context to coordinate traffic across connected corridors. To address this trade-off, FairLight introduces an intermediate regional abstraction between network-level coordination and local signal control.

The framework consists of four principal components: **(i) automatic regional construction, (ii) a network-level Meta-Policy, (iii) intersection-level Sub-Policies, and (iv) a safety-constrained signal execution layer**. Regional construction transforms the signalized road network into coherent traffic regions, the Meta-Policy extracts global spatial-temporal information and generates network-level control targets, and the Sub-Policies translate this global guidance into local phase decisions using intersection and neighborhood information. The resulting hierarchy is trained jointly using PPO with an adversarial goal-alignment mechanism.

3.5 Traffic Network Representation and Automatic Regionalization

Manually decomposing a large urban road network into meaningful control regions is impractical because the number and connectivity of signalized intersections can vary substantially across networks. FairLight therefore represents the signalized road network as a weighted graph

$$G = (V, E), \tag{2}$$

where each vertex $v_i \in V$ represents a signalized intersection and each edge $e_{ij} \in E$ represents the connectivity between two signalized intersections. Non-signalized junctions between traffic lights are contracted during graph construction so that the resulting graph directly represents the controllable signalized-intersection topology.

The connectivity between intersections is characterized using a traffic-dependent edge weight rather than purely static topology. For an edge e, the implemented weighting function is

$$w_e = \alpha n_e + \beta \omega_e + \gamma \kappa_e n_e, \tag{3}$$

where n_e denotes the number of vehicles on the edge, ω_e represents accumulated waiting time, and

$$\kappa_e = \mathrm{clip}\left(1 - \frac{v_e}{v_e^{\max}}, 0, 1\right) \tag{4}$$

denotes the congestion ratio. The implementation uses $\alpha = 1.0$, $\beta = 0.3$, and $\gamma = 2.0$. These weights are averaged over a 1000-step simulation window, allowing the regional structure to reflect persistent traffic interactions rather than a transient traffic state.

The resulting weighted graph is partitioned into communities using graph community-detection methods. In particular, Louvain and Leiden identify communities by maximizing graph modularity,

$$Q = \frac{1}{2m} \sum_{i,j} \left(A_{ij} - \frac{k_i k_j}{2m}\right) \delta(c_i, c_j), \tag{5}$$

where A_{ij} is the weighted adjacency matrix, k_i and k_j are the weighted degrees of vertices i and j, m is the total edge weight, and $\delta(c_i, c_j)$ indicates whether the two vertices belong to the same community. DBSCAN and hybrid graph-density approaches are additionally supported, allowing regionalization itself to be evaluated as an experimental variable rather than treated as a fixed preprocessing choice.

This regionalization produces a set of regions

$$\mathcal{R} = \{R_1, R_2, \ldots, R_M\}, \tag{6}$$

which forms the intermediate abstraction used by the hierarchical controller. Each region provides aggregated traffic information to the Meta-Policy, while individual intersections within the regions remain independently controllable by the Sub-Policies.

3.6 Hierarchical Meta-Policy

The Meta-Policy provides network-level coordination by observing the temporal evolution of traffic conditions across the generated regions. For each region R_k, the regional state is represented as

$$s_k^t = \left[N_k^t, W_k^t, x_k, y_k\right] \in \mathbb{R}^4, \tag{7}$$

where N_k^t denotes the aggregate stopped-vehicle count, W_k^t denotes aggregate waiting time, and (x_k, y_k) represents the normalized spatial centroid of the region.

Because traffic conditions are inherently temporal, the Meta-Policy maintains a sliding window of $T = 20$ regional observations. A Transformer encoder processes the regional tokens jointly to capture spatial dependencies between regions. A learnable global token is introduced to obtain a network-level representation

$$F_g \in \mathbb{R}^4. \tag{8}$$

The temporal representations of individual regions are subsequently processed using an LSTM. The resulting hidden states are concatenated across the M regions and transformed through an MLP to obtain a global representation $G \in \mathbb{R}^{16}$. The Meta-Policy then generates two network-level control targets,

$$(G_w, G_q) = \mathrm{Linear}_{16\to 2}(G), \tag{9}$$

where G_w represents the desired global waiting-time level and G_q represents the desired global queue-length level.

Thus, rather than directly selecting signal phases throughout the network, the Meta-Policy establishes a global traffic-control objective that can be pursued by the lower-level intersection agents. This separation allows long-range traffic coordination to be handled at a compressed regional level while preserving local responsiveness at individual intersections.

3.7 Intersection-Level Sub-Policy

The Sub-Policy converts global guidance into local signal-control actions. At each intersection i, the local observation is represented by

$$o_i^t = [v_i, q_i, o_i, f_i, s_i, w_i, v_i^{avg}, p_i, c_i, d_i] \in \mathbb{R}^{10}, \tag{10}$$

where the components capture local traffic volume, queue length, occupancy, flow, stops, waiting time, speed, pressure, congestion, and delay. The observations are normalized using an exponential moving average and clipped to a bounded range before being supplied to the policy network.

A local multilayer perceptron first encodes the intersection state. Since traffic conditions at one intersection are influenced by neighboring intersections, the encoded representation is augmented using a Graph Attention Concat (GAC) mechanism over the intersection-adjacency graph. For a neighboring pair (i, j), the attention coefficient is computed as

$$e_{ij} = \text{LeakyReLU}\left(a^T[h_i \| h_j]\right), \qquad \alpha_{ij} = \text{softmax}_j(e_{ij}), \tag{11}$$

and the resulting neighborhood representation is constructed through concatenation,

$$z_i = h_i \| \alpha_{i1} h_{j_1} \| \cdots \| \alpha_{i4} h_{j_4}. \tag{12}$$

The neighborhood representation is then combined with the global Meta-Policy embedding:

$$s_i^t = [z_i \| F_g]\,. \tag{13}$$

This representation simultaneously provides the Sub-Policy with **local traffic information, neighboring traffic context, and network-level coordination guidance**.

A shared actor-critic network processes this representation and produces a probability distribution over signal phases. Because intersections may have different numbers of valid signal phases, invalid action indices are masked before the final softmax operation. Parameter sharing across intersections allows the same learned policy structure to be deployed across heterogeneous signalized intersections while retaining local observations and graph-based neighborhood information.

3.8 Joint Hierarchical Optimization

The Meta-Policy and Sub-Policies are optimized jointly. The Sub-Policy is trained using Proximal Policy Optimization (PPO) with Generalized Advantage Estimation (GAE). The temporal-difference residual is

$$\delta_t = r_t + \gamma V(s_{t+1})(1 - d_t) - V(s_t), \tag{14}$$

and the generalized advantage estimate is

$$A_t = \delta_t + \gamma\lambda(1 - d_t)A_{t+1}, \tag{15}$$

with $\gamma = 0.99$ and $\lambda = 0.95$. The PPO policy objective uses the clipped probability ratio

$$L_{\text{policy}} = -\min\left(r_t A_t, \text{clip}(r_t, 1-\epsilon, 1+\epsilon)A_t\right), \tag{16}$$

where $\epsilon = 0.2$.

The hierarchical relationship is enforced through an additional goal-alignment term. Let W_{global} and Q_{global} denote the actual network-wide waiting time and queue length. The Meta-Policy is trained using

$$L_{\text{Meta}} = \text{MSE}((G_w, G_q), (W_{\text{global}}, Q_{\text{global}})) + \eta_1 r_g, \tag{17}$$

while the Sub-Policy receives an alignment objective encouraging the realized network state to follow the Meta-Policy's targets:

$$L_{\text{Sub}} = L_{\text{AC}} + \eta_2\left[\beta_q(Q_{\text{global}} - G_q) + \beta_w(W_{\text{global}} - G_w)\right]. \tag{18}$$

Here, $\eta_1 = \eta_2 = 0.1$ and $\beta_q = \beta_w = 0.5$. The alignment gradient is detached from the Meta-Policy in the Sub-Policy loss, preventing the lower-level policy from directly modifying the higher-level representation through this term.

The local reward combines intersection-level traffic efficiency with the global goal-alignment reward:

$$r_i^t = -\left(q_i^t + w_i^t + d_i^t + p_i^t - s_i^t\right), \tag{19}$$

$$r_g^t = -\left[\beta_q(W_{\text{global}} - G_q) + \beta_w(Q_{\text{global}} - G_w)\right], \tag{20}$$

and

$$r^t = r_i^t + r_g^t. \tag{21}$$

This formulation ensures that each intersection is evaluated according to its own local traffic conditions while simultaneously receiving a network-level coordination signal.

3.9 Safety-Constrained Signal Execution

The learned policy does not directly bypass traffic-signal safety constraints. Every requested phase transition is passed through a deterministic execution layer that enforces a minimum green interval followed by a yellow transition before the new green phase is activated.

Specifically, a requested phase change is permitted only after the current phase has satisfied the minimum-green requirement. The controller then enters a yellow transition for the prescribed duration before committing the requested green phase. In the current implementation, the minimum green time is 5 simulated seconds and the yellow transition lasts 3 seconds.

This separation between **learned decision-making and deterministic safety enforcement** prevents the reinforcement-learning agent from producing physically invalid or unrealistically rapid signal transitions.

3.10 End-to-End Control Process

At every reinforcement-learning decision step, the framework first aggregates the current traffic state for each automatically generated region and updates the temporal regional-state buffer. The Meta-Policy then processes this temporal representation to obtain the global embedding and network-level queue and waiting-time targets. Simultaneously, each intersection obtains its normalized local observation, incorporates information from neighboring intersections through graph attention, and fuses this representation with the global Meta-Policy embedding.

The resulting Sub-Policy actions are passed through the safety-constrained signal execution layer before being applied to the traffic simulator. The environment then advances the traffic simulation, computes local and global rewards, and stores the resulting transition for policy optimization. Meta-Policy updates are performed periodically, while Sub-Policy PPO updates are performed after the rollout buffer reaches the specified horizon.

The complete architecture can therefore be expressed as the following hierarchical mapping:

$$\begin{aligned} G \to \mathcal{R} \to \text{Meta-Policy} \to F_g, (G_w, G_q) \\ \to \text{Sub-Policies} \to \text{Signal Actions} \to \text{Traffic Environment}. \end{aligned} \tag{22}$$

This design separates **regional coordination, temporal network reasoning, local spatial interaction, and physical signal execution** into distinct layers. Consequently, the framework can automatically adapt its regional abstraction to different road networks while retaining a shared local control mechanism and a global coordination signal.

4 Multi-Objective Reward Function Design

An effective traffic signal controller must optimize multiple objectives simultaneously rather than maximizing traffic throughput alone. Conventional reinforcement learning approaches primarily optimize queue lengths

or average waiting times, often producing policies that systematically disadvantage certain intersections, neglect pedestrian requirements, and fail to prioritize emergency vehicles adequately. To address these limitations, we design a multi-objective reward function that jointly optimizes traffic efficiency, fairness, and emergency vehicle priority.

The overall reward at each decision step is defined as

$$R_{\text{total}} = \frac{4}{9}R_{\text{eff}} + \frac{2}{9}R_{\text{fair}} + \frac{3}{9}R_{\text{emergency}}, \tag{23}$$

where each component corresponds to a distinct traffic management objective.

4.1 Traffic Efficiency Reward

The efficiency component encourages continuous improvement in network performance by rewarding reductions in congestion rather than rewarding only low congestion states. It is defined as

$$R_{\text{eff}} = -\beta_1 \Delta Q - \beta_2 \Delta W + L_{\text{avg}}, \tag{24}$$

where ΔQ denotes the change in total queue length between successive decision steps, ΔW represents the change in cumulative vehicle waiting time, and L_{avg} is the average local throughput reward aggregated across all intersections. The coefficients β_1 and β_2 determine the relative importance assigned to queue reduction and waiting-time reduction.

Unlike conventional reward formulations based solely on absolute queue lengths, the use of temporal differences provides denser learning signals by rewarding measurable improvements between consecutive time steps. This enables faster convergence while encouraging policies that continuously reduce congestion.

4.2 Fairness Reward

Conventional traffic-signal controllers commonly optimize aggregate measures such as queue length, average delay, and throughput. These objectives can improve network efficiency but do not guarantee equitable service to individual vehicles: a movement repeatedly deprioritized by the controller may experience prolonged waiting even when network-level metrics improve. FairLight therefore treats fairness as a complement to traffic efficiency and emergency-vehicle responsiveness, consistent with the multi-objective reward defined above.

Vehicle waiting-time monitoring. For each vehicle v and intersection j, let $w_{v,j}(t)$ denote its accumulated waiting time at time t. The timer increases by one simulation step whenever the vehicle is stopped (speed below 0.1 m/s) on an incoming lane of j, is kept while the vehicle creeps forward in the queue, and is discarded when the vehicle is served and leaves the incoming lanes of j; a new waiting episode then starts at the next intersection. The same definition is used in training and evaluation. A vehicle is flagged as a starvation risk when

$$w_{v,j}(t) \geq W_{\max}, \qquad W_{\max} = 90 \text{ s}. \tag{25}$$

Reaching $W_{\max}$ flags the vehicle and increases its consideration in subsequent phase selection: the requested phase of the Sub-Policy is replaced by a feasible phase that gives a green to the lane of the longest-waiting flagged vehicle (the current phase is kept if it already does). The 90-second threshold is a design threshold and not a universal guarantee that every vehicle is served within 90 seconds: demand, conflicting movements, incidents, clearance intervals, and physical feasibility may make the threshold temporarily or persistently unattainable. The override only changes the *requested* phase; the controller never bypasses the minimum-green requirement (5 steps) or the 3 s yellow clearance of Section 3.9, and no override is issued during a yellow transition.

To align this safeguard with the reward, define the local maximum waiting time at intersection j as $W_j(t) = \max_{v \in \mathcal{V}_j(t)} w_{v,j}(t)$, where $\mathcal{V}_j(t)$ is the set of tracked vehicles currently waiting at that intersection. The threshold penalty is then $\max(0, W_j(t) - W_{\max})$. The maximum is taken locally per intersection, not over the whole episode or network.

Intersection-level fairness using Jain's index. At each intersection, Jain's Fairness Index (JFI) summarizes how evenly a set of non-negative service scores is distributed. For n included vehicles with scores $x_i \geq 0$,

$$J = \frac{(\sum_{i=1}^{n} x_i)^2}{n \sum_{i=1}^{n} x_i^2}. \tag{26}$$

For non-zero scores, $J \in [1/n, 1]$, and a value closer to 1 indicates a more even distribution. If all scores are zero the denominator vanishes; we adopt the convention $J = 1$ (no disparity) and, when no vehicle is waiting at all, set every fairness term to zero.

The choice of x_i matters. Raw waiting time should not be used as a service score, because a higher waiting time indicates *worse* service, and equal but very high waiting times still yield $J = 1$. We therefore use the bounded, non-negative score that is inversely related to waiting time,

$$x_i = \frac{1}{1 + w_{i,j}/W_{\text{ref}}}, \tag{27}$$

where $W_{\text{ref}} = 30$ s is a fixed reference time chosen before training. The score, the population (all vehicles waiting on the incoming lanes of j), the sampling interval (every simulation step), and the handling of vehicles not yet served are identical across methods and experiments. JFI measures equality, not absolute performance or starvation by itself, so it is reported alongside the mean, maximum, and 95th-percentile waiting time, queue length, throughput, and emergency-vehicle response time, and it is computed locally per intersection before any city-wide aggregation so that local disparities are not concealed by a network average.

Fairness reward. Let W_j be the maximum waiting time at intersection j as defined above, let $\sigma^2_{\hat{w},j}$ be the variance of the normalized waiting times $\hat{w}_i = w_{i,j}/W_{\text{ref}}$ over the local population, and let J_j be the JFI of the scores x_i. The dimensionless local fairness penalty is

$$P_{\text{fair},j} = \lambda_{\text{var}} \operatorname{Var}(\hat{w}) + \lambda_{\text{starve}} \max\Big(0, \frac{W_j - W_{\max}}{W_{\max}}\Big) + \lambda_J (1 - J_j), \tag{28}$$

and the fairness reward is $R_{\text{fair},j} = -P_{\text{fair},j}$, which is added to the local reward r_j^t of Section 3.8. Normalizing waiting times by W_{ref} and the starvation excess by $W_{\max}$ is important because waiting-time variance has squared-time units whereas the starvation and JFI terms are dimensionless. The coefficients $\lambda_{\text{var}}, \lambda_{\text{starve}}, \lambda_J \geq 0$ are hyperparameters; larger disparity, threshold excess, or inequality reduces the reward. Their initial values $(\lambda_{\text{var}}, \lambda_{\text{starve}}, \lambda_J) = (1, 10, 10)$ were chosen so that the fairness terms are not negligible next to the queue and waiting terms of r_j^t; they are design choices, not empirically established optimal values. The earlier coefficients 0.3, 0.25 and 0.15 applied to quantities with incompatible units and are superseded by this formulation. Envy and pedestrian terms are not part of the implemented fairness reward.

4.3 Emergency Vehicle Reward

Emergency vehicles require substantially higher priority than regular traffic because delays directly affect emergency response times. Their reward component is therefore defined as

$$R_{\text{emergency}} = -W_{\text{emergency}}, \tag{29}$$

where $W_{\text{emergency}}$ is the cumulative waiting time experienced by emergency vehicles. This formulation directly penalizes any delay encountered by ambulances, fire trucks, or police vehicles, encouraging the controller to allocate green phases immediately upon emergency vehicle detection.

Finally, the combined reward is normalized using an exponential moving average before being supplied to the reinforcement learning algorithm. Reward normalization reduces reward variance, improves numerical stability during training, and accelerates policy convergence.

4.4 Evaluation Protocol

Fairness is evaluated jointly with efficiency and safety. At minimum we report the mean, maximum, and 95th-percentile vehicle waiting time; the proportion of vehicles exceeding 90 seconds; intersection-level JFI; queue length; throughput; and emergency response time. The controller is compared against the same controller without fairness terms, under identical traffic demand, random seeds where applicable, signal constraints, and simulation horizons, and multiple runs are reported with variability estimates. A reduction in average delay alone is insufficient evidence of improved fairness, and a higher JFI alone is insufficient unless absolute delays remain acceptable. Emergency pre-emption intentionally gives some vehicles higher urgency, so results distinguish ordinary-vehicle fairness from emergency-vehicle responsiveness.

5 Future Works

5.1 Enhanced Emergency Vehicle Routing

The routing mechanism presented in this work improves on EMVLight's reactive dynamic Dijkstra heuristic by integrating route planning into the hierarchical Meta-Policy. However, several directions remain open. At each Meta-Policy decision step, a candidate route set is generated from current regional pressure readings and the known destination, and sub-goals are issued to Sub-Policy agents along candidate corridors to reduce queue lengths before EMV arrival. This transforms routing from a two-horizon problem (current link and next link) into a multi-horizon planning problem aligned with the hierarchical structure. Future work will extend this to settings where the destination is unknown at dispatch time—for example, in mass-casualty events where the nearest available hospital is itself dynamically determined—and to scenarios with more than two simultaneously active EMVs whose corridors intersect repeatedly. The Louvain community structure provides a natural abstraction for corridor planning at the boundary level: because inter-community transitions represent sparse bottleneck links, the Meta-Policy can reason about sequences of community transitions rather than individual links, substantially reducing the planning horizon while preserving global routing awareness.

5.2 Envy-Free Traffic Control

The envy-free regularization introduced in this work operates within individual Louvain communities during training. Future work will investigate adaptive scheduling of λ_e across training: starting with a high efficiency weight and gradually increasing the fairness weight as the policy matures may accelerate convergence while preserving equity in the final policy. Additionally, the definition of envy used here is pairwise and local; extending it to group envy—where a set of intersections collectively prefers another group's allocation—would capture more subtle structural inequities in large networks. The interaction between the envy-free penalty and the EMV pre-emption reward also warrants further study: during active EMV passage, pre-emption agents and normal agents necessarily receive asymmetric rewards, and calibrating λ_e dynamically based on EMV proximity could allow the system to temporarily suspend fairness constraints along the active corridor while preserving them network-wide.

5.3 Accident-Based Traffic Control on Real City Maps

The accident injection model used in this work assigns a fixed severity parameter κ to each incident and samples accident locations according to a crash probability distribution derived from historical report data. Future work will model accidents with evolving severity: an initial partial blockage ($\kappa < 1$) that transitions to complete blockage ($\kappa = 1$) as secondary vehicles arrive and then gradually clears as the incident is resolved. This temporal evolution of supply disruption is a qualitatively harder problem for the Meta-Policy, requiring it to track incident lifecycle stages rather than treating each accident as a static constraint. Evaluation on the Coimbatore city map will be extended to multi-district scenarios in which accidents in one part of the city cascade congestion into adjacent zones, stressing the inter-community coordination capacity of the Meta-Policy. The combination of accident-induced rerouting, multiple active EMVs, and envy-free fairness enforcement represents the most realistic and demanding test scenario for the complete framework, and constitutes the primary experimental target for the next phase of this research.

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